

Thomas Ngo

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Portfolio | LinkedIn | GitHub

EDUCATION

University of Alberta

Bachelor of Science: Computing Science with Honors

Edmonton, AB

Sep 2023 – Expected June 2027

- **GPA: 3.86 / 4.00**
- **Honors & Awards:** International Admission Scholarship, Regional Excellence Scholarship.
- **Relevant Coursework:** Machine Learning I & II, Algorithms, Visual Recognition (Computer Vision), Web Applications & Architecture, Search & Planning in AI, Mathematical Statistics, Numerical Methods.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, C/C++, C#, SQL, R, LaTeX, HTML/CSS.
- **AI & Data Science:** Deep Learning, Natural Language Processing, Computer Vision (OpenCV), LLM Integration, Scikit-Learn, PyTorch, NumPy, Pandas, Time Series Analysis.
- **Frameworks & Web:** React, Node.js, Express, Spring Boot, VueJS, Tailwind CSS, JUnit.
- **Cloud & DevOps:** Docker, Kubernetes, Git, CI/CD Pipelines, Firebase, Linux/Unix, GCP.
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, SQLite3, Firestore.

PROFESSIONAL EXPERIENCE

Viettel Military Industry and Telecoms Group

Software Engineering Intern (Web Programming)

Remote / Hanoi, Vietnam

May 2026 – July 2026

- Completed intensive training in enterprise-scale system architecture and full-stack development.
- Engineered and deployed containerized web modules using **Spring Boot** and **VueJS**, utilizing **Docker** and **Kubernetes** to ensure scalable and stable service delivery.
- Optimized backend performance by implementing caching layers with **Redis** and streamlining **MySQL** queries, significantly improving data retrieval speeds for high-traffic environments.
- Applied **Agile/Scrum** methodologies to the software development lifecycle, collaborating with senior mentors to pitch technical innovations to executive leadership.

SELECTED PROJECTS

Android Event Lottery Application | Java, Firebase, CI/CD

Oct 2025 – Dec 2025

- Designed a real-time event management system for 500+ potential users; utilized **Firestore** for live data synchronization and integrated QR-code check-in functionality.
- Architected automated **CI/CD pipelines** with JUnit testing, ensuring code reliability and reducing manual deployment overhead by 30%.

Web-Based Document Editor | React, Node.js, PostgreSQL, JWT

June 2025

- Built a high-performance text editor with asynchronous document state management and a secure **RESTful API**.
- Implemented robust user authentication via **JWT**, ensuring secure data persistence and session management.

Social Media Application | Python, SQLite

Nov 2024

- Developed a full-featured CLI platform with secure database operations and protection against SQL injection.
- Designed complex relational schemas to manage many-to-many relationships for user engagement and content tracking.

Optimizer Animated Visualizer | React, TypeScript, Python, Node.js, Railway

May 2026

- Developed a full-stack benchmarking platform to simulate and visualize 12 ML optimizer-kernel combinations; engineered a **Python/NumPy** engine to execute high-performance numerical optimization experiments.
- Built an interactive **React/TypeScript** dashboard featuring real-time playback controls and dynamic animations using **Recharts**, enabling granular analysis of learning curve convergence.
- Designed a modular **RESTful API** to bridge the numerical backend with the frontend, deploying the system via **Railway** with an automated CI/CD pipeline for rapid iteration.

EXTRACURRICULAR & LEADERSHIP

UAlberta Book Club | *VP of Marketing*

May 2025 – April 2026

- Directed social media strategy and promotional campaigns, leading to a measurable increase in club engagement.

UofA Computing Science Department | *Student Volunteer*

Oct 2024

- Represented the Computing Science department, engaging with prospective students to provide guidance and answer inquiries about undergraduate programs, highlighting opportunities within the faculty.